MILITARY SPECIFICATION

DISPLAY UNIT, TELETYPEWRITER TT-581/AG AND MOUNTING MT-4088/AG

THIS SPECIFICATION HAS BEEN APPROVED BY

THE NAVAL AIR SYSTEMS COMMAND, DEPARTMENT OF THE NAVY

1.0 SCOPE

1.1 Scope - The equipment covered by this specification shall be an Alphanumeric Display device intended for use in transmitting, receiving, storing and editing teletypewriter messages.

1.2 Classification - The equipment covered by this specification shall consist of the following items:

ITEM	TYPE DESIGNATION	APPLICABLE PARAGRAPH
Teletypewriter Display Unit	TT-581/AG	3.5
Mounting	MT-4088/AG	3.5.5

1.3 Associated Equipment - This equipment shall operate with TT-264/AG Teletypewriter and CV-1053/ARC Signal Data Converter.

2.0 APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-C-172	Case, Base, Mounting, and Mounts Vibration for use with Electronic Equipment in Aircraft
MIL-W-5088	Wiring; Aircraft, Installation of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment

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MIL-P-7788	Panels, Information Integrally Illuminated
MIL-M-7793	Meter, Time Totalizing
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-T-18303	Test Procedures; Preproduction and Acceptance for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-M-81288	Mounting Bases, Flexible Form Plastic

Naval Air Systems Command

AR-5	Microelectronic Devices Used in Avionics Equipment, Procedures for Selection and Approval of
AR-8	Versatile Avionic Shop Test System, Avionic System Compatibility General Requirements for
AR-9	Vast Test Programs, General Requirements for
AR-10	Maintainability of Avionics Equipment and Systems, General Requirements for

Military

MIL-STD-188	Military Standard, Military Communication System Technical Standards
MIL-STD-461	Electromagnetic Interference Characteristics, Requirements for
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement for

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MIL-STD-463	Definitions and Systems of Units, Electromagnetic Interference Technology
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-781	Reliability Tests, Exponetial Distribution
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MS17322	Meter, Time Totalizing, 400 cycle, 115 Volt Rev. B
FED-STD-222	Radiation Standard for Communications and Other Information - Processing Equipment

2.1.1 Availability of Documents

(1) When requesting specifications, standards, drawings, and publications, refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

3. REQUIREMENTS

3.1 Preproduction - The specification makes provision for preproduction testing.

3.2 Parts and Materials - In the selection of parts and materials fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

(1) Microelectronic technology shall be considered and microelectronic items shall conform to requirements specified herein.

(2) Other parts and materials requirements shall conform to Specification MIL-E-5400.

(3) Nonrepair subassemblies, as outlined in Specification MIL-E-5400, shall be used when practicable. The general

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size of the subassembly and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable. (See 6.5)

(4) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.2.1 Nonstandard Parts and Materials Approval - Approval for the use of nonstandard parts and materials (including electron tubes, transistors and diodes) other than microelectronic devices shall be obtained as outlined in Specification MIL-E-5400. Microelectronic devices shall be approved as outlined in AR-5.

3.2.2 Microelectronic Modular Assemblies - When used, Microelectronic modular Assemblies shall meet the requirements of Specification AR-5.

3.2.3 Modules - The electric portions of the equipment shall be modularized in accordance with Specification AR-10.

3.3 Design and Construction - The equipment shall conform with all the applicable requirements of Specification MIL-E-5400 for design, construction and workmanship, except as otherwise specified herein.

3.3.1 Total Weight - The total weight of the equipment including mounting, excluding cables, shall be minimum consistent with good design and shall not exceed 46 pounds.

3.3.2 Reliability -

3.3.2.1 Operational Stability - The equipment shall operate with satisfactory performance, continuously or intermittently for a period of at least 500 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use under standard conditions.

3.3.2.2. Operating Life - The equipment shall have a total operating life of greater than 10 years with reasonable servicing and replacement of parts. Parts requiring scheduled replacement shall be specified by the contractor.

3.3.2.3 Reliability in Mean Time Between Failure (MTBF) - The equipment shall have 2000 hours of mean (operating) time between failures when tested and accepted as outlined under the requirements of 4.4.3.

3.3.2.4 Time Totalizing Meter - The following unit shall contain a time totalizing meter in accordance with Specification MIL-M-7793.

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UNIT	TYPE OF METER
Teletypewriter Display Unit TT-581/AG	MS17322

3.3.3 Cabling and Connections -

3.3.3.1 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-E-5400.

3.3.3.2 Interconnection Cabling - The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-W-5088. The external wiring shall be unshielded except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the equipment.

3.3.4 Panel Illumination - The panel illumination shall comply with Specification MIL-P-7788. The primary source of power shall be 5.0 volts and shall be supplied by the aircraft.

3.3.5 Interchangeability - The equipment shall meet the interchangeability requirements of Specification MIL-E-5400.

3.3.6 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall meet the requirements of MIL-STD-461, MIL-STD-462, and MIL-STD-463.

3.3.7 Provisions for Maintainability - Built-in test equipment, construction and packaging, and provisions for test points shall be as specified in Specification AR-10.

3.3.7.1 Compatibility with VAST - The equipment shall be compatible with the Versatile Avionic Shop Test System (VAST) and shall meet the requirements of Specification AR-8. When required, VAST Test Programs shall be furnished in accordance with Specification AR-9. If VAST Test Programs exist for the equipment, and changes to the equipment are made which will affect the fault diagnosis procedure, changes to the existing Test Programs shall be prepared as part of the equipment changes in accordance with Specification AR-9.

3.3.8 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307.

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3.3.9 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient (25° C ± 5° C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input power voltage	115 ± 1 VAC, 400 Hz, Three-Phase

3.3.10 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in Specification MIL-E-5400 for Class 1A equipment, except as modified herein:

3.3.10.1 Vibration - The equipment shall operate satisfactorily when subjected to the vibration requirements of Curves II and IV of Specification MIL-E-5400.

3.3.11 Warm-Up Time - The time required for the equipment to warm up prior to operation shall be kept to a minimum and shall not exceed 2 minutes under standard conditions and 5 minutes at extreme service conditions.

3.3.12 Input Electrical Power -

3.3.12.1 Operating Power - The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance when energized from the following power sources with characteristics and limits as defined in MIL-STD-704. The power required shall not exceed the specified amounts.

(1) AC Power (Three-Phase) 115 Category "B", 100VA

3.3.12.2 Lighting Power - Input power for lighting shall require not more than 0.1 amp at 5 volts.

3.3.12.3 Transient Protection: The display unit shall contain circuitry necessary for protection from any input power transients as defined in MIL-STD-704. Such protection shall prevent loss of data from the memory as a result of power transients and the only loss of data allowed will be that data which would have been supplied by external sources during power interruption. When normal power is restored after a power interruption, the equipment shall resume operation within 500 msec.

3.3.13 Standby Provisions - No provisions shall be made for applying power to the CRT filament only.

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3.4 Performance - Unless otherwise specified, values set forth to establish the requirements of satisfactory performance apply to performance under both standard and extreme service and input power conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.4.1 Operation - The equipment is intended for use as a message composing editing, correcting and storage device. It will accept input messages from a teletypewriter keyboard or from a radio receiving line via a CV-1053/ARC Signal Data Converter. It will also, upon operator command, transmit stored messages via an RF transmitter or to an associated page printer. All messages to and from the equipment shall be selectable either in the International Telegraph Alphabet No. 2 American Version or the American Standard Code for Information Interchange (ASCII) as specified in MIL-STD-188B; paragraph 2.7.3. Function switches shall be provided to aid the operator in composing, editing, correcting, controlling, storaging and transmitting messages.

3.5 Detail Requirements - The teletypewriter Display Unit shall meet the following requirements.

3.5.1 Function - The Display Unit shall operate with a TT-264/AG Teletypewriter and CV-1053/ARC Signal Data Converter or equivalent, to transmit and receive teletype messages. It shall store, display, and provide the capability to edit these messages prior to transmissions.

3.5.2 Form Factor - The Display Unit shall be enclosed in a case conforming to Specification MIL-C-172 having dimensions not greater than the following: 20 1/2 inches long by 7 1/2 inches wide by 7 5/8 inches high.

3.5.3 Weight - The weight of the Display Unit, less shock mounting, shall not exceed 42 pounds. This weight shall include all subassemblies necessary for display generation and control and all support assemblies such as power supplies.

3.5.4 Contents - The Display Unit shall contain the following major functional components which shall provide the following performance characteristics:

3.5.4.1 Storage - The Display Unit shall contain memory for storage of at least 4096 6-bit data words or 8-bit data words. A control shall be provided to select the type of data word to be stored. In the send mode, data shall be regenerated into storage. Capability for destruction of the stored data words shall be provided.

3.5.4.2 Symbol Generator - The Display Unit shall contain a symbol generator capable of operating on a 7 by 9 matrix. The symbol generator shall be the dot matrix type and all symbols shall be displayed on a 5 by 7 matrix. Two row or two column intersymbol spacing shall be provided by the remainder of the 7 by 9 matrix.

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3.5.4.3 Display Characteristics - The Display Unit shall provide the following display format options and capabilities.

3.5.4.3.1 Display Format - The Display Unit shall be capable of displaying alpha-numeric symbols on 14 lines per display page with an additional line provided for page indication. The maximum number of pages displayed shall be eight but the actual number may be limited by the fixed bit storage capability. A function switch shall be provided for mode selection to operate with:

(1) The International Telegraph Alphabet No. 2, American version or
(2) The American Standard Code for Information Interchange (ASCII).

In the first mode of operation, a maximum of 35 characters per line shall be displayed at the maximum eight page capacity. In the second or ASCII mode of operation a maximum of 40 characters per line shall be displayed with not less than seven pages being displayed.

3.5.4.3.2 Refresh Rates - The display shall be refreshed every 30.0 milliseconds or less.

3.5.4.3.3 CRT Characteristics - The CRT display viewing area shall be a minimum of 4 inches vertically and 4 inches horizontally. The CRT phosphor shall be P31 or equivalent. Minimum brightness shall be adjustable from zero to 30-foot-lamberts at 30-foot-candles ambient illumination. Minimum contrast ratio shall be 8:1 at 30-foot-candles of ambient illumination. Spot size shall be adjustable from 10 to 20 mils to provide for maximum character definition.

3.5.4.3.4 CRT Character Positioning - The CRT character positioning shall be as follows: -

(1) The minimum character size shall be a 10 point character and shall not vary more than ±1.0% in height or width between any two successive presentations.
(2) The vertical and horizontal position of any two successive presentations of the same character shall not vary more than ±2%.
(3) On any display page, variations in character dimensions or in horizontal spacings between characters, due to location on the page, shall not exceed 15%.
(4) Vertical (or horizontal) position of any symbol shall not vary more than 15% from a theoretical line through the center of the row (or column) in which the symbol is contained.
(5) Spacing between two consecutive rows or columns shall not vary more than 10.0%. Total accumulative error shall not exceed 1/16".

3.5.4.4 Interface Requirements - The Display Unit shall be capable of interfacing to the TT-264/AG Teletypewriter and the CV-1053/ARC Signal Data Converter or equivalents.

3.5.4.4.1 Interface Data Format - The character code format shall be as specified in MIL-STD-188. The code set shall be International Telegraph Alphabet No. 2 Communication Set and 64 symbol subset American Standard Code for Information Interchange (ASCII).

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3.5.4.4.2 Electrical Interface Requirements - The electrical interface shall be compatible with the following equipment requirements.

(A) Teletypewriter Set

(1) High Current range (20 to 80 MA) 115 ohms resistive at 600 MA.

(2) Low Current range (1 to 5 MA) 2200 ohms resistive at 5 MA.

(3) Transmission Signal Source Impedance shall not exceed 5.0 ohms.

(4) Type of signals - Neutral (non-polarized, 10 to 80 MA with mark pulses as current and space pulses as no current).

(B) Signal Data Converter

(1) Input Impedance 1100 ohms

(2) Pulse output signal to display unit shall be closing and opening, with respect of converter relay output contacts with a rating of 15 watts for resistive loads at a maximum of 125 volts or 0.12 ampere.

(3) Transmit mode operation-Pulse Input signal from Teletypewriter Display Unit shall be grounded line for converter mark output; open line or positive voltage (up to +200 volts for converter space output).

(C) Special options for auxillary equipment

(1) Input - 80 microamperes at 2 VDC for mark pulses, no current for space pulses.

(2) Output Neutral - 20 or 60 milliamperes at -40 VDC for mark pulses, no current for space pulses.

(3) Output Polar: +20 or +30 milliamperes for mark pulses- -20 or -30 milliamperes for space pulses.

3.5.4.4.3 Interface Timing Requirements - The interface shall operate at data transfer rates of 45.45, 50, 74.2 and 150 bauds at 5 and 7 level rate codes.

3.5.4.4.4 Operator Controls - The Display Unit when interfaced with the TT-264/AG teletypewriter or other local keyboard/printer device

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shall provide operator control of message entry and editing. Several function switches shall be provided on the front panel of the display. Other semifixed adjustable controls for the display intensity; horizontal and vertical deflection linearity; gain and etc., shall be readily accessible when a cover is removed from the case. If other than hand tools are required for adjustment, such tools shall be furnished and clipped to the unit inside the case.

TABLE 1. OPERATOR CONTROLS

NAME	TYPE	FUNCTION
POWER ON	Switch alternate Action Indicator	Turns power to the display on and off.
MEM CLEAR (Memory Clear)	Switch momentary action Indicator	Removes all symbol codes from storage.
PAGE CLEAR	Switch momentary action Indicator	Removes all symbols from the screen from the cursor position
LINE CLEAR	Switch momentary action Indicator	Removes all symbols to right of cursor in the line containing the cursor.
RESET	Switch momentary action Indicator	Transfer the cursor to the first symbol position (upper left corner) of that page.
PAGE ADV (Page Advance)	Switch momentary action Indicator	Selects sequentially one of eight pages from memory to be displayed. The cursor is transferred to the first symbol position of the selected page.
SEND	Switch momentary action Indicator	The message is transmitted from the cursor to the end of message.
LINE UP	Switch momentary action Indicator	Transfer the cursor to the same symbol position of the next line up.
LINE DOWN	Switch momentary action Indicator	Transfer cursor to the same symbol position of the next line down.

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TABLE 1. OPERATOR CONTROLS (CONT)

NAME	TYPE	FUNCTION
BKSP (Backspace)	Switch momentary action Indicator	Transfer the cursor one symbol position to the left at a continuous rate not to exceed 3 characters per second.
SKIP	Switch momentary action Indicator	Transfer the cursor one symbol position to the right.
INSERT	Switch alternate action Indicator	Permits insertion of new message with destruction of old message.

3.5.4.5 Electrical Connections - Connections to external circuits shall be provided as follows:

Reference Designation	Receptacle Type	Function
J1	Fed std. 222	Power Input
J2	Approved	Input/output to Peripheral Equipment

3.5.5 Mounting MT - 4088/AG - The teletypewriter display unit shall be capable of being installed on a plane surface within an aircraft or vehicle. A multi-functional barrier vibration isolator of a flexible plastic type shall be employed subject to approval of the Naval Air Systems Command, The mounting shall conform to the applicable requirements of MIL-M-81288. The weight of the mounting shall not exceed 4 pounds.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements.

(1) Preproduction (First Article) Tests

(2) Initial Production Tests

(3) Acceptance Tests

(4) Life Tests

4.2 Preproduction (First Article) Tests - Preproduction tests shall be made on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall be accomplished under the approved test procedure of 4.6. The Government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the government.

4.2.1 Preproduction (First Article) Test Data - The contractor shall submit all data collected in conducting these tests to the procuring activity for review.

4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with the procedures of Specification MIL-T-5422 and interference tests in accordance with MID-STD-461, 462 and 463.

4.2.3 Preproduction (First Article) Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. The approval preproduction sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered as one of the equipments under the contract.

4.2.4 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of non-compliance with the above shall

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constitute cause for rejection and for equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Initial Production Tests - One of the first ten production equipments shall be selected and sent at the contractor's expense to a designated Government laboratory for tests. This equipment shall be selected by the procuring activity after the equipment has successfully passed all individual tests. The preproduction sample shall not be selected for this test.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance and quality and that it meets all applicable requirements.

4.3.2 Accessory Material - In addition to the complete equipment submitted for Initial Production Tests the contractor shall also submit such necessary material and data necessary to test the equipment.

4.3.3 Initial Production Sample Approval - Approval of the initial Production Sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the Initial Production Sample to pass any of the tests shall be cause for deliveries under the contract to cease until proper corrective action is approved and accomplished. Correction action shall also be accomplsihed on equipment previously accepted when requested by the procuring activity.

4.3.4 Reconditioning of Initial Production Test Sample - On completion of the initial production test the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4,4. Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the Government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish tests reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacturer shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

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(2) Sampling Tests

(3) Reliability Assurance Tests

(4) Special Tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Manufacturing Run in Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment terperature to stabilize and to check sufficient charactersitics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing Run in Test - Each equipment shall be operated under the conditions specified herein for a period of 10 hours without failures. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment. This test shall be deleted if the reliability test includes a test on each equipment which consumes at least 10 hours of operation.

Temperature	Ambient Room
Humidity	Ambient Room
Vibration	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ± 3 g's

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 10 hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship.

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Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plans for 5 minutes and lateral to the plans for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions, the vertical direction shall be used. During the 10 hour period of operation following the 10 minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10 minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action should be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 10 hour period specified above may be composed of two 5 hour periods to conform with standard working hours.

4.4.2 Sampling Tests - Equipments selected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling tests by the Government inspector in accordance with the following: (Sampling tests shall be conducted only when Reliability Assurance Tests, of 4.4.3 are deleted by contractual action).

Quantity of Equipments Offered for Acceptance	Quantity to be Selected For Sampling Test
First 10	0*
Next 50	1
Next 75	1
Next 100	1

*NOTE: If by contract action the Initial Production Test (4.3) is deleted, a Sample Test shall be conducted on one equipment from the first 10 produced.

4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, initial production, individual and special tests.

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(3) Manufacturing run in test specified in 4.4.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipment.

4.4.3 Reliability Assurance Tests - Reliability Assurance Tests will be conducted using MIL-STD-781. Tests as required by both the Qualification Phase and the Sampling Phase shall be conducted.

4.4.3.1 Qualification Phase - Prior to the acceptance of equipments under the contract or order, a minimum of three (3) equipments shall be tested as outlined in MIL-STD-781, under the section entitled "Qualification Phase of Production Reliability Tests". The maximum number of equipments to be used shall be those listed in Table 5 of MIL-STD-781. For the qualification Phase, Test Level E shall be used. The Accept-Reject Criteria for Test Plan IV shall be used.

4.4.3.2 Reliability Sampling Phase Tests - Samples of the production equipment shall be tested as outlined in MIL-STD-781 (as modified herein), under the Section entitled, "Sampling Phase of Production Reliability Tests". For the sampling Phase (or All-Equipment Test), Test Level E shall be used.

4.4.3.2.1 Lot Size for Sampling Phase - The equipment constructed during one month shall be one lot. The Accept-Reject criteria for Test Plan IV shall be used to determine the length of the tests in each lot. Unless otherwise specified in the contract or order, no equipments shall be shipped until an Accept decision is reached under each lot.

4.4.3.3 Test Details - The test details such as the length of the test cycle, the length of the heat portion of the cycle, the performance characteristics to be measured, special failure criteria, preventive maintenance to be allowed during the test, etc., shall be part of the test procedures to be submitted and approved by the procuring activity prior to the beginning of the Qualification Test Phase of the Reliability Assurance Tests.

4.4.4 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling or reliability assurance tests.

4.4.4.1 Special Test Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early equipment after an engineering or material change.

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(2) Whenever failure reports or other information indicate that additional tests are required. (This will be determined by the procuring activity).

4.4.4.2 Scope of Tests - Special Tests shall consists of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedures and submit it to the procuring activity for approval prior to conducting the tests.

4.4.5 Equipment Failure - Should a failure occur during either the sampling, reliability assurance or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirements until reasonable assurance is obtained that the defect has been satsifactorily corrected.

4.5 Life Test - The contractor shall furnish all samples and shall be responsible for accomplishing the life tests. The test shall be of 300 hours duration and shall be conducted on equipments that have passed the individual test. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the Government inspector in accordance with the following. (Equipments which have successfully passed the Initial Production Test, Sampling Tests, Reliability Tests, or Special Tests may be selected for life tests.) When reliability tests are conducted, the life test shall be omitted if, during the reliability tests, a quantity of equipments equal to, or more than, that listed below receive at least 300 hours each of test time,

Quantity of Equipments Offered for Acceptance	Quantity to be Selected For Life Test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof.

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4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions:

Temperature	Normal room
Altitude	Normal ground (0-5000 ft.)
Humidity	Room ambient
A.C. voltage	115 ± 5 volts (at lower applicable frequency)

4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases or operation.

4.5.3 Performance Check - At approximately 8 hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

(1) Part type number

(2) The circuit reference symbol number

(3) The part function

(4) Name of the manufacturer

(5) Nature of the failure

(6) The number of hours which the part operated prior to failure

4.5.4.1 Failure Report - In the event of a failure, the Government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or materil corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

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4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the Government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts, such procedures will be approved and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including tests, when deemed necessary.

4.7 Reconditioning of Tested Equipment - Equipment which has been subjected to initial acceptance and life tests shall be reconditioned by the contractor by replacing all wear or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.8 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.9 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Government inspector.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specifications MIL-E-17555, and MIL-STD-794. In the event the equipment is not covered in Specification MIL-E-17555, the method of preservation for Level A shall be determined in accordance with the selection chart in Appendix D of MIL-STD-794.

6. NOTES

6.1 Intended Use - This equipment is intended to interface with airborne teletypewriter equipment and shall be used in lieu of mechanical tape perforation and tape reader equipment.

6.2 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(1) Title, number and data of this specification

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(2) Selection of applicable levels of packaging and packing (See 5.1)

6.3 Precedence of Documents - When the requirements of contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.4 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accomplished by complete supporting information.

6.5 Non-Repairable Subassemblies - As a general rule, non-repairable subassemblies should be encapsulated or hermetically-sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high voltage power supplies), the non-repairable subassembly should evidence a Mean-Time-to-Failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

6.6 Type Designations - The parentheses (*) when used in the equipment type designation, will be deleted or replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.3.8. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

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